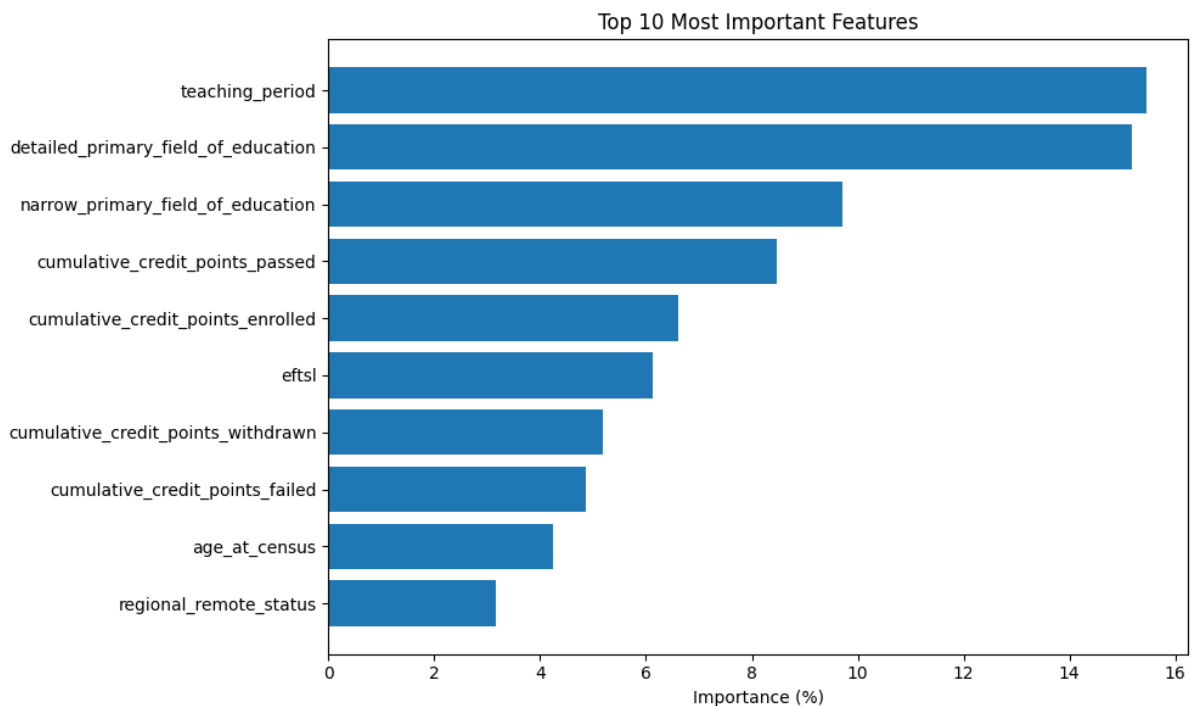


Feature Importance

The machine learning code in **Machine Learning Model.ipynb** extracts feature importance information as well. It includes a plot of the top 10 important features and their percentage importance as such:



Usually, random forests randomly split trees which leads to every run producing different results and thus, different importance rankings. However, the random forest classifier used for this task has a fixed seed parameter. Therefore, the importance ranking remains unchanged between runs. The results above reveals that the machine learning model that is run on the synthetic data identified that teaching period is the largest factor that contributes to student attrition, followed mostly by student enrolment or academic history information such as study area, credit points and EFTSL. It appears that the lower ranking features are demographic information such as age and regional/remote status.

While enrolment and academic history information being the largest factors contributing to student attrition makes sense, teaching period being the biggest factor seems a little unexpected. This result may be flawed due to the defect mentioned in **T115 Synthetic Data Validation Documentation.pdf** where teaching periods with the same year and teaching period type have different start and end dates – which does not reflect real teaching period information.